ALCHEMY AOE ALLIANCE

TYPE: HANDBOOK (MTHD)		**PROJECT: N/A**
MTHD--007		**REVISION: X1**
Structure, Single Division Stack League		**Page 1 of 8**

Table of Contents

1 - Referenced Documents

This “Method” does not depend on any components.

2 - Definitions

3 - Background

Creation of this specification was part of an initiative to modularize many recurrent elements of Alchemy League handbooks, minimizing reading required for returning players.

4 - Overview

This specification explains matchmaking for a unique type of Round Robin tournament, intended as an informative reference for handbooks using this system.

5 - Play Structure

Figure 5 below illustrates the core League design:

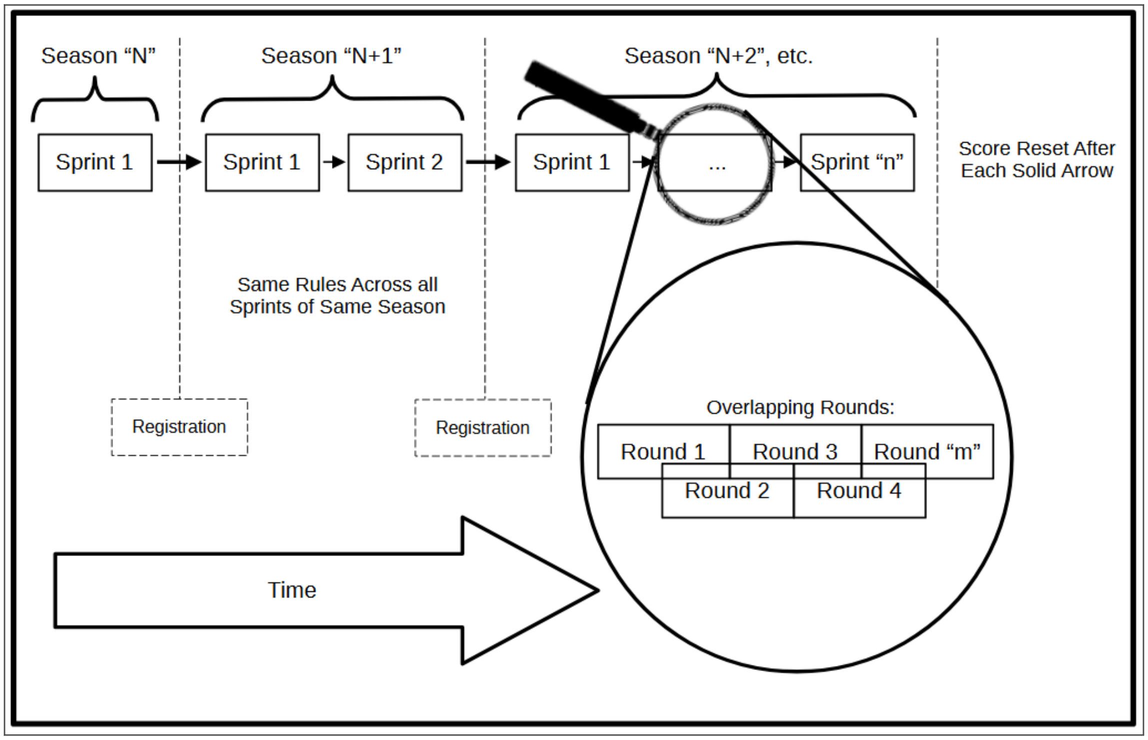


Figure 5: League Structure

The Season begins shortly after its registration closes and ends after a predetermined number "n" of Sprints (defined in handbook) are completed.

Each Sprint consists of a predetermined quantity "m" of overlapping Rounds.

The League assigns each participant a different opponent to play each Round.

Scoring and compensation are calculated after aggregating results from all Rounds within the same Sprint, and reset when the next begins.

There is no elimination from Round to Round, but the Season handbook may specify performance required to play in subsequent Sprints.

5.A - Exceptions

An ideal setup for the League is described in §5, but because the League must accommodate any number of participants, some minor exceptions are required:

5.A.1 - Round Wild

Although most league players will complete one match per Round, it is impossible for some participant counts. An extra round –known as “Round Wild” — is allocated for any matches that are left over. Its features are:

- Only one “Round Wild” per Sprint.
- “Round Wild” begins and ends with its corresponding Sprint. Matched players may play *any time* within that period.
- Assignment of any set of matched players to “Round Wild” is random.

5.A.2 - Repeated Opponent

Although rare, occasional repeats of lower seed match-ups are required so that each league participant has one match for each planned Round.

5.A.3 - Technical Impossibility

Total player count must be **greater than** two times the number of Rounds, plus 1.

If the tournament fails to achieve the threshold sign-ups to support planned quantity of Rounds, then improvisation is necessary. To accommodate low sign-up count, the number of Rounds can be reduced or broken into smaller pieces where possible (E. G. two Sprints of 3 instead of one Sprint of 6).

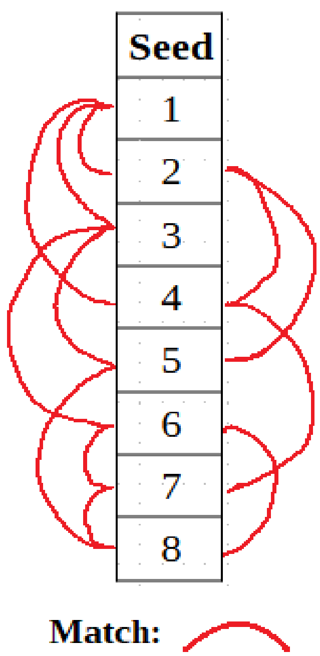
6 - Matchmaking

6.A - Stack

Alchemy League is a single-division Round Robin, but with the special condition that players are matched against a unique subset of all participants. The matching depends on the total participant count and the number of Rounds in the Sprint, with the only requirements being:

- Each player gets the same number of matches
- Each player is matched with closest available opponents

Figure 6.A illustrates an example for Sprints consisting of three rounds, each with eight players:



Matchup
1 – 2
1 – 3
1 – 4
2 – 4
2 – 5
3 – 5
3 – 6
4 – 7
5 – 8
6 – 7
6 – 8
7 – 8

Figure 6.A: Example Matchmaking Scenario

To generate the left array of Figure 6.A, skill levels are calculated for all eight participants, and sorted in descending order, with highest skill set to seed “1” and lowest skill set to seed “8”. Then, curves are drawn, linking each seed to three others in what is referred to as “the stack”.

These connections are selected carefully by computer algorithm to ensure that each integer or “seed number” is paired with exactly three others, biasing for closer neighbors. The general strategy involves starting at the top and bottom of the stack, and reaching inward in an increasingly staggered manner. Since middle seeds are in demand for matches from **both** top and bottom, halving the density of that demand ensures that each seed is given the same number of pairings. For this reason, middle seeds will never pair with nearest neighbors.

The right array of Figure 6.A summarizes all connections scrawled on the left. Each integer may be counted to confirm the same number of appearances across both sides of the dash, in this case a frequency of three.

Figure 6.B represents a single example (combination of 8 and 3), among infinitely many possibilities outside the limitation set in §5.A.3.

6.B - Distribution

Once matches are decided, they are distributed across Rounds in the tournament under the following rules:

- Each player is tasked to complete only one set per Round.
- Rounds quantity to be minimized; no more than one “Round Wild”.

Following from the example in §6.A, Figure 6.B shows two potential distributions of matches:

	Round 1	Round 2	Round 3	Round Wild
Not Ideal	1 – 2 3 – 5 7 – 8	1 – 3 2 – 4 5 – 8 6 – 7	1 – 4 2 – 5 3 – 6	4 – 7 6 – 8
Ideal	1 – 2 3 – 5 4 – 7 6 – 8	1 – 3 2 – 4 5 – 8 6 – 7	1 – 4 2 – 5 3 – 6 7 – 8	

Figure 6.B: Potential Distributions of 8 to 3 Combination

As the number of players and particularly number of Rounds grows, the opportunity for non-ideal distribution also increases, requiring a computer computer algorithm to organize.

Signatures:

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